Perturbation	Original text	Perturbed text
Character-level		
Insertion	Who was the first governor of Alaska?	Who was the firsdt governor of Alaska?
Deletion	Mercury, what year was it discovered?	Mercury, what year was it discovred?
Replacement	Who is the Prime Minister of Canada?	Who is the Prime Monister of Canada?
Swapping	What is the primary language in Iceland?	What is the primary lnaguage in Iceland?
Repetition	How many hearts does an octopus have?	How many heartts does an octopus have?
CMW	What kind of gas is in a fluorescent bulb?	What kind of gas is in a florescent bulb?
LCC	How many hearts does an octopus have?	How many hearts does an OCTOPUS have?
Word-level		
Deletion	How much was a ticket for the Titanic?	How much a ticket for the Titanic?
Repetition	What is another name for vitamin B1?	What is another name name for vitamin B1?
RWS	What precious stone is a form of pure carbon?	What valued rock is a form of pure carbon?
Negation	What planet is known as the "red" planet?	What planet is not known as the "red" planet?
SPV	What does a barometer measure?	What do a barometer measure?
Verb tense	Why in tennis are zero points called love?	Why in tennis were zero points called love?
Word order	What is the most common eye color?	What is the common most color eye?

Table 2: Character-level and word-level perturbation examples from the TREC question classification dataset. CMW: Common Misspelled Words, LCC: Letter Case Changing, RWS: Replacement With Synonyms, SPV: Singular/Plural Verbs.

Almost all the character-level perturbations presented here were already tested in adversarial attack scenarios (Heigold et al., 2018; Zeng et al., 2020; Zhang et al., 2020), but were not yet implemented in a non-adversarial testing framework, except the misspelling perturbation implemented by CheckList. Among the word-level perturbations, *Deletion*, *Repetition*, *Singular/plural verbs*, *Word order*, and *Verb tense* were not already used to test the robustness. However, *Negation* was included in CheckList, and *Replacement with Synonyms* was used for adversarial attack (Dong et al., 2020; Ren et al., 2019).

4.1 Character-level perturbation

These perturbation methods randomly select a word, denoted as $Word_i$, and apply perturbations to its characters. They are described in the following.

Insertion. A character is randomly selected and inserted in a random position (except the first and last position) if $Word_i$ contains at least three characters.

Deletion. A character is randomly selected and deleted if $Word_i$ contains at least three characters. The last and first characters of $Word_i$ are never deleted.

Replacement. A character is randomly selected and is replaced by an adjacent character on the keyboard.

Swapping. A character is randomly selected and swapped with the adjacent right or left character in $Word_i$.

Repetition. A character in a random position (except the first and last position) is selected and a copy of it is inserted right after the selected character.

Common misspelled words. If a word in the input text appears in the Wikipedia corpus of common misspelled words[4], it is replaced by its misspelling.

Letter case changing toggles the letter case, i.e. converts a lower case character to its upper case form and vice versa. The letter case changing is done for either the first or all the characters of $Word_i$. The type of letter case changing is specified in a random manner.

4.2 Word-level perturbation

Deletion randomly selects a word from the input sample and removes it.

Repetition selects a random word, makes a copy of it, and inserts it right after the selected word.

Replacement with synonyms replaces words contained in the sample by their synonyms extracted from the WordNet lexical database (Miller, 1995).

Negation. It identifies verbs in the sample, then injects negations by converting positive verbs to negative, or removes negation by converting negative verbs to positive. The goal is to

[4] https://en.wikipedia.org/wiki/Wikipedia:Lists_of_common_misspellings/For_machines/

Task	LM	Test set	Character-level perturbation methods						
			Insertion	Deletion	Replace	Swap	Repeat	CMW	LCC
TC	BERT	90.4	77.4	76.2	76.1	76.5	78.8	58.4	78.3
	RoBERTa	**93.1**	79.2	**78.9**	76.3	**76.7**	**80.8**	60.5	78.9
	XLNet	92.0	78.1	78.3	**76.5**	75.2	80.2	61.5	77.4
	ELMo	84.8	**80.4**	78.5	74.7	75.6	79.6	**61.9**	**80.8**
SA	BERT	92.2	77.1	75.6	75.5	**78.3**	77.9	62.0	76.7
	RoBERTa	**94.0**	79.3	76.8	75.1	76.2	79.3	64.0	78.3
	XLNet	93.1	78.3	78.7	75.9	73.8	**81.1**	**65.6**	78.9
	ELMo	87.6	**79.7**	**79.0**	**76.2**	78.1	78.4	64.1	**79.1**
NER	BERT	92.6	83.6	80.7	81.4	82.5	81.9	71.3	81.2
	RoBERTa	**93.3**	**84.3**	80.9	**81.7**	**83.1**	**82.4**	**71.8**	**81.5**
	XLNet	92.7	83.9	**81.1**	81.3	82.7	81.5	71.6	81.3
	ELMo	90.2	83.0	80.2	80.9	82.2	81.3	70.8	81.0
SS	BERT	82.5	72.9	71.5	73.0	74.3	74.2	68.6	73.8
	RoBERTa	**83.9**	**73.5**	**72.8**	**73.6**	**75.1**	**74.7**	**69.5**	**74.9**
	XLNet	83.3	73.3	72.0	73.2	74.6	74.1	67.9	74.4
	ELMo	80.7	71.1	70.9	72.3	73.8	72.5	67.0	72.6
QA	BERT	91.6	82.7	80.5	81.1	81.9	79.8	68.6	80.7
	RoBERTa	**94.9**	**84.1**	**81.7**	**82.9**	**83.2**	81.6	**72.5**	**84.0**
	XLNet	93.4	83.5	81.1	82.3	81.9	**82.8**	71.5	83.3
	ELMo	85.5	80.6	79.5	76.0	78.3	80.1	67.9	81.1

Table 3: Performance of the language models on the test sets and character-level perturbed samples of the downstream tasks. For every task and every perturbation method, the highest score is shown in bold face. CMW: Common Misspelled Words, LCC: Letter Case Changing.

investigate the ability of the NLP system in adapting its outcome to reflect the injected or removed negation.

This perturbation method operates based on a set of rules that assess verbs, subjects, and verb tenses based on POS tags, then applies an appropriate rule to construct the test sample. For example, if the POS tag of a verb is VBZ, the verb appears in the third person simple present form. Therefore, the verb is replace by [does not + VBP] where VBP is the basic form of the verb, in order to inject negation into the sample.

Singular/plural verbs. It simulates a common error in real use cases, i.e. using plural form of a verb instead of the singular form, and vice versa, usually with a third-person subject. This perturbation does not usually change the text's meaning in most NLP tasks if the task does not rely on the subject-verb agreement. Therefore the NLP system should treat the perturbed sample as an unperturbed text.

Word order. It randomly selects M consecutive words from the sample and changes the order in which they appear in the text. The goal is to investigate whether the NLP system is sensitive to word ordering or it only decides based on the presence of words in the input.

Verb tense. It converts present simple or continuous verbs to their corresponding past simple or continuous forms, or vice versa. The goal is to assess the sensitivity of the NLP system to changing the verb tense in tasks where the verb tense is not important to the output. In this case, the system's output should not change after modifying the verb tense. This method first extracts POS tags to identify verbs and their subjects. It then converts the verb tense using the `mlconjug3` package and reconstruct the sentence with the new verb tense.

5 Experimental results

All the experiments were performed on a computer with an Intel Core i5-9600K CPU at 3.70GHz, 32 GB of RAM, and a GeForce RTX 2080 Ti graphic card (GPU) with 11 GB dedicated memory. Perturbation methods ran on CPU; fine-tuning on training sets, and evaluating on test sets and perturbed samples ran on GPU.

5.1 Performance on perturbed inputs

Since it has been proven that sentences that contain few typos, misspellings, or minor character-level errors can be still fully understandable to humans (Belinkov and Bisk, 2018; Xu and Du, 2020), character-level perturbations are not expected to change the text's meaning in most cases. Therefore, they can be automatically produced and used for testing the robustness of NLP systems.

On the other hand, some word-level perturbations may change the text's meaning.

			Word-level perturbation methods						
Task	**LM**	**Test set**	**Deletion**	**Repeat**	**RWS**	**Negation**	**SPV**	**VT**	**WO**
TC	BERT	90.4	75.1	**89.3**	65.7	89.1	88.2	89.0	74.5
	RoBERTa	**93.1**	**76.2**	88.7	73.2	**90.3**	**89.5**	89.4	78.5
	XLNet	92.0	**76.2**	87.5	72.7	89.4	89.0	**89.6**	**83.1**
	ELMo	84.8	72.9	82.8	**75.1**	83.5	83.6	81.2	62.9
SA	BERT	92.2	73.7	87.6	67.5	**84.6**	88.2	**90.1**	76.4
	RoBERTa	**94.0**	74.5	**90.1**	**74.2**	83.9	**88.7**	88.6	77.5
	XLNet	93.1	**74.7**	88.5	74.1	82.3	88.6	89.3	**83.8**
	ELMo	87.6	72.0	80.6	73.1	75.4	84.6	82.9	65.9
NER	BERT	92.6	81.4	83.1	74.1	85.3	88.2	89.1	70.7
	RoBERTa	**93.3**	**82.3**	**83.9**	**74.5**	**85.8**	**88.6**	**89.4**	71.1
	XLNet	92.7	81.9	83.7	73.9	85.6	88.3	88.7	**74.8**
	ELMo	90.2	79.7	82.1	69.3	82.4	85.1	84.9	68.5
SS	BERT	82.5	72.6	74.1	69.4	68.5	75.2	75.6	72.0
	RoBERTa	**83.9**	**74.1**	**74.8**	**70.0**	**69.2**	75.7	**76.7**	73.9
	XLNet	83.3	73.3	74.5	69.8	68.7	**75.8**	76.2	**75.3**
	ELMo	80.7	69.8	71.7	67.4	66.0	73.2	72.8	72.6
QA	BERT	91.6	78.4	**89.6**	71.5	84.9	88.0	90.3	76.7
	RoBERTa	**94.9**	**79.9**	89.3	**78.1**	**86.5**	**89.7**	**91.2**	79.0
	XLNet	93.4	79.2	89.5	77.3	86.1	89.1	90.9	**85.8**
	ELMo	85.5	73.5	81.4	75.0	82.7	84.1	81.9	67.3

Table 4: Performance of the language models on the test sets and word-level perturbed samples of the downstream tasks. For every task and every perturbation method, the highest score is shown in bold face. For three perturbation methods, i.e. *Deletion*, *Negation*, and *RWS*, 200 perturbed samples were used in the experiments. RWS: Replacement With Synonyms, SPV: Singular/Plural Verbs, VT: Verb Tense, WO: Word Order.

Consequently, the perturbed samples should be monitored to make sure they are still meaningful with respect to the NLP task at hand, and are consistent with the original label in the dataset. Otherwise, they should not be used for testing the robustness, or the label should be changed to reflect the change and preserve the consistency.

We separately applied every character-level perturbation method to all test samples in a dataset, and all the resulting perturbed samples were used to evaluate the robustness of the language models. A hyperparameter named Perturbation Per Sample (PPS) specified the maximum number of perturbations in a sample.

We monitored and filtered perturbed samples resulted from three word-level perturbations that may change the text's meaning. These perturbations are *Deletion*, *Negation*, and *Replacement with synonym*. For every sample whose meaning was changed by these three methods, and a change in the test set label was necessary to preserve consistency, we altered the label if it was applicable. If a proper label could not be assigned to the perturbed sample or the resulting text was no longer meaningful, we excluded the sample from the evaluations. Since monitoring and filtering every single perturbed sample was extremely time-consuming (such that approximately one minute was needed on average to check the meaningfulness of a perturbed sample and its consistency with the test set label), we corrected labels and filtered perturbed samples for the above three methods until 200 samples were collected for every dataset; then we used these samples to evaluate the models on perturbed inputs. We performed this manual curation of perturbed samples for all values of PPS that we experimented, i.e. values in the range [1, 4]. Appendix B presents the number of perturbed samples checked in the manual curation procedure until reaching 200 test samples for every dataset and different values of PPS. The manual curation was performed by three annotators who had sufficient English language knowledge to properly judge about the meaningfulness and consistency of perturbed samples.

Since the rest of word-level perturbations are not expected to change the text's meaning with respect to the NLP tasks in our experiments, we did not monitor and filter them; they were produced and used automatically. Again, the PPS hyperparameter controlled the maximum number of perturbations in every sample.

Table 3 and Table 4 present the performance of the language models on character-level and word-level perturbed samples, respectively. These results are reported for PPS=1. The performance of the language models on original, unperturbed test sets